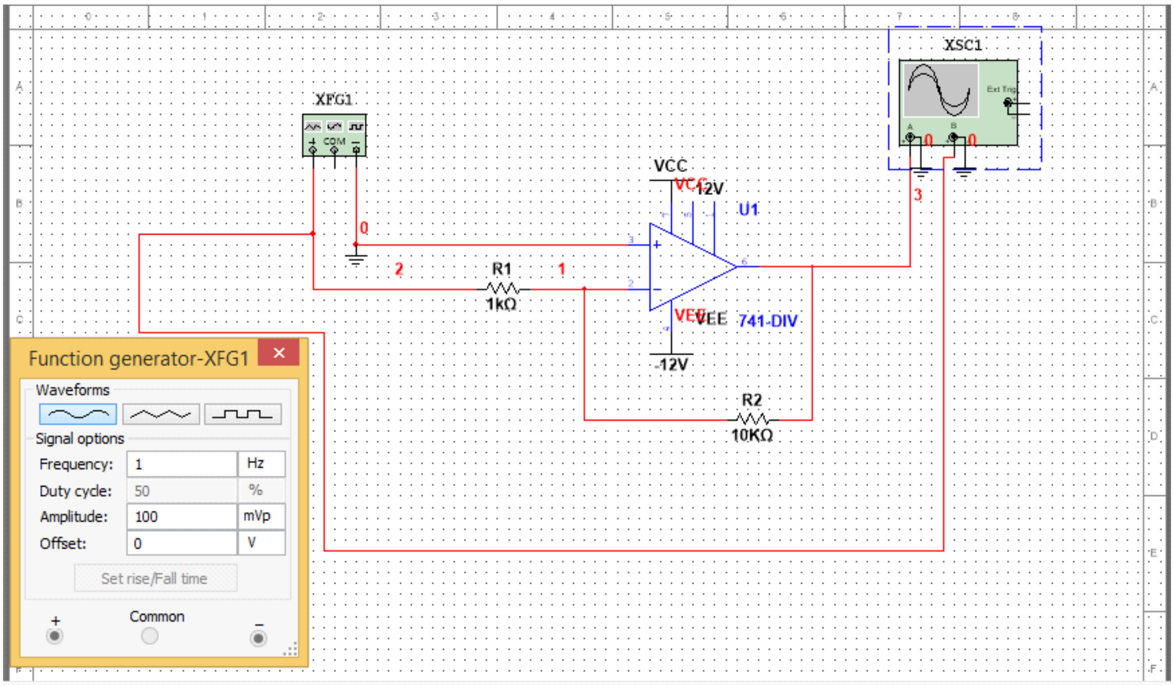
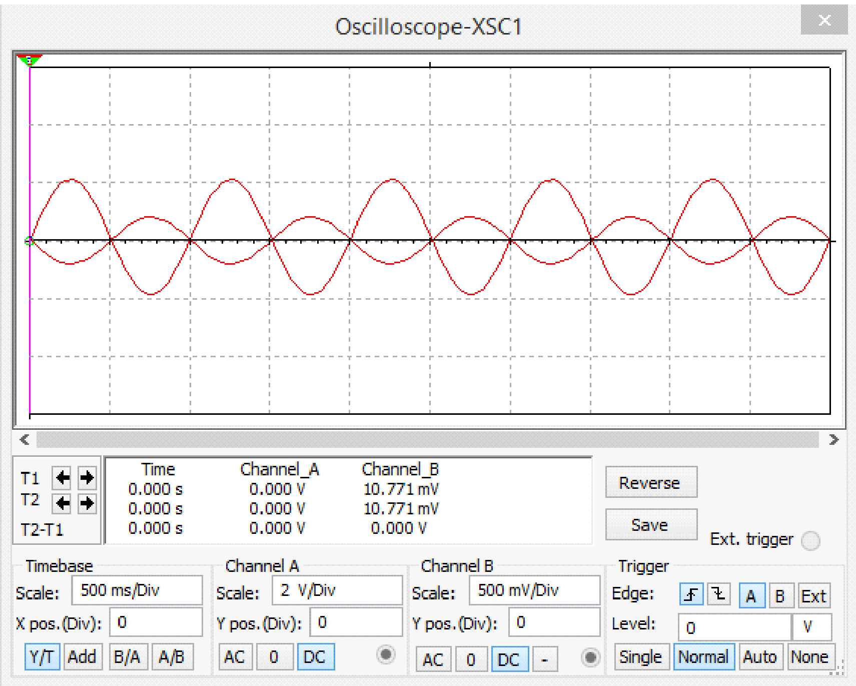
EXPERIMENT 1A: INVERTING AMPLIFER

AIM: Study and simulation of inverting amplifier and calculate the gain using MULTISIM software worksheet

COMPONENTS: 1K (R1) RESISTOR, 10 K (RF) RESISTORS, 741 IC (OP-AMP), FUNCTION GEN, OSCILLOSCOPE



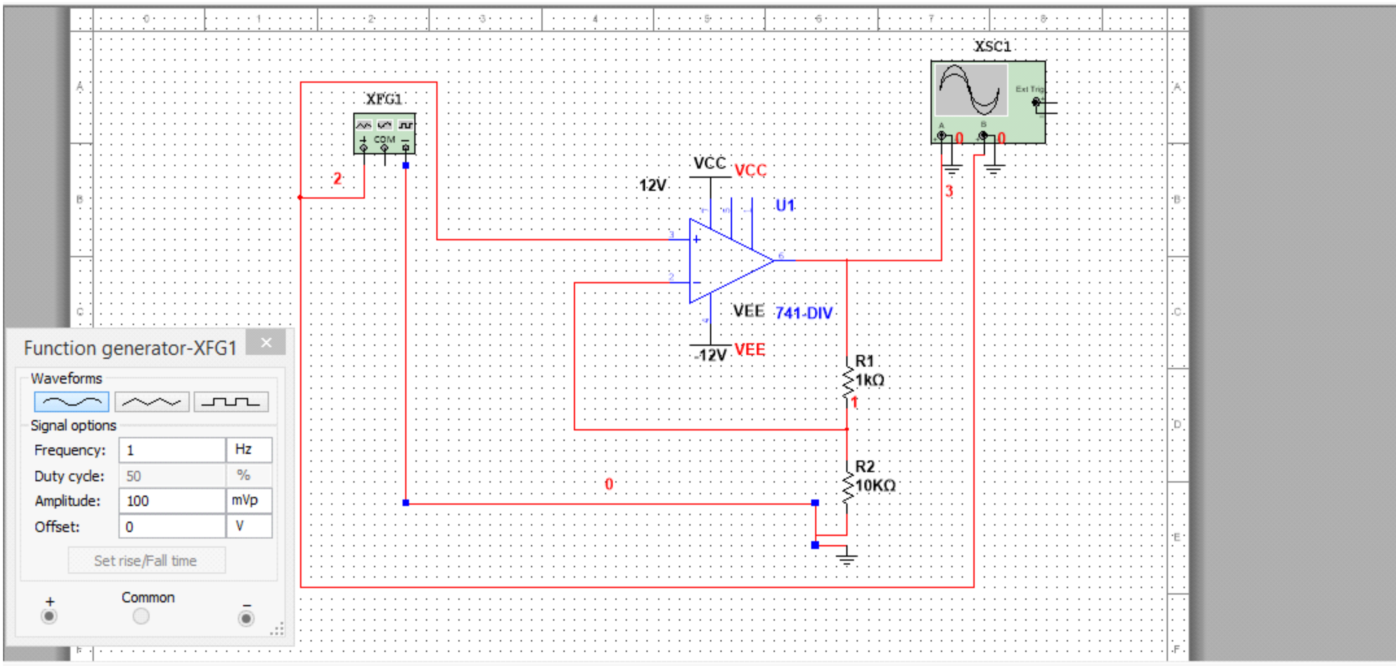


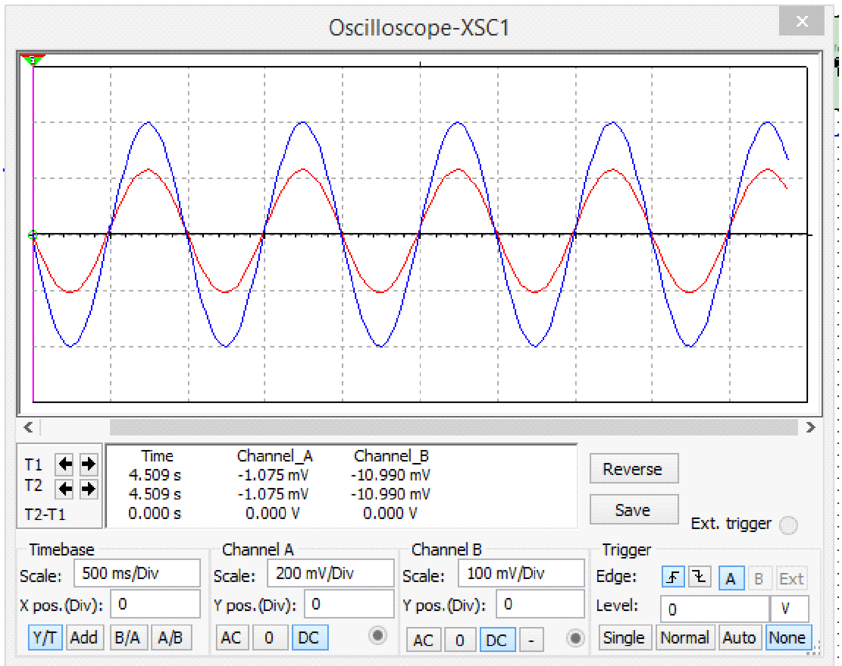
gain= -vo/vin =-10/1=-10

EXPERIMENT 1: NON INVERTING AMPLIFER

AIM: Study and simulation of non-inverting amplifier and calculate the gain using MULTISIM software worksheet

COMPONENTS: 1K (R1) RESISTOR, 10 K (RF) RESISTORS, 741 IC (OP-AMP), FUNCTION GEN, OSCILLOSCOPE

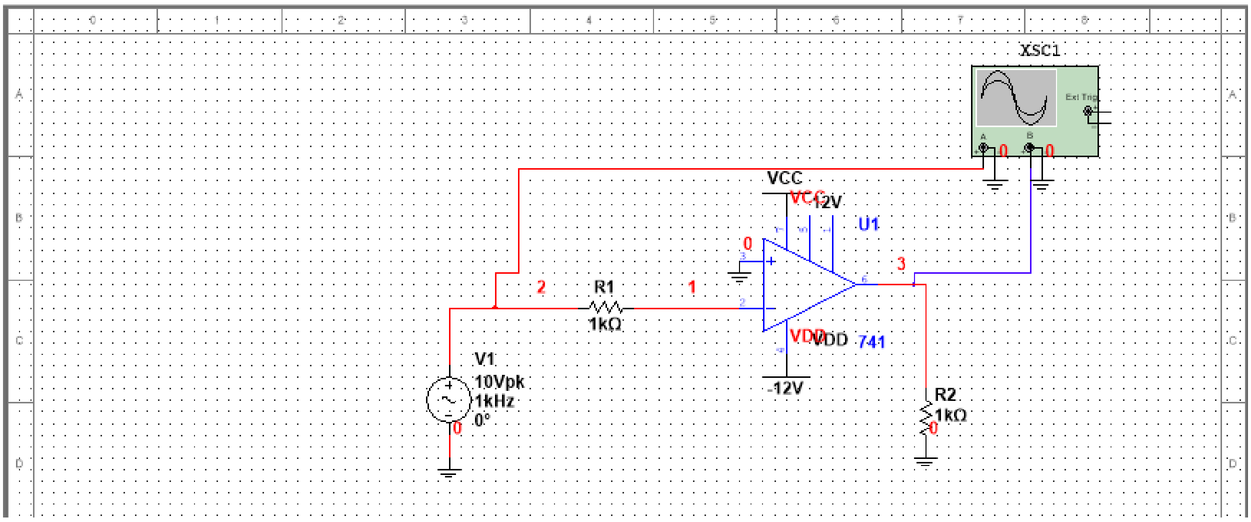


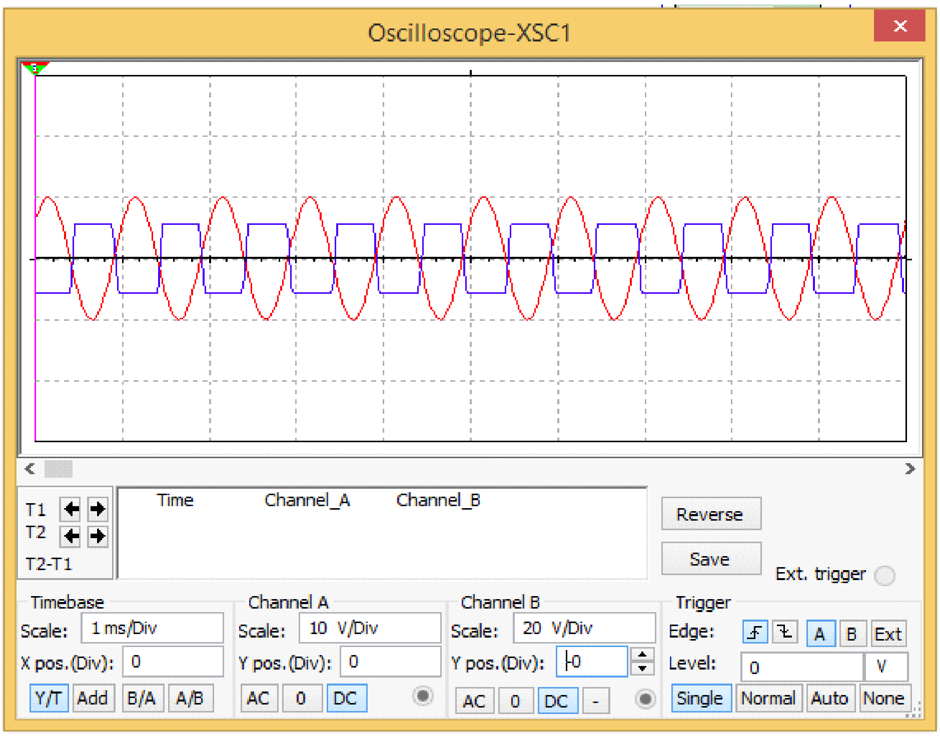


Experiment 2A: zero crossing detector

AIM: Study and simulation of zero crossing detector and observing the wave form using MULTISIM software worksheet

COMPONENTS: 1K (R1) RESISTOR, 1K (R) RESISTORS, 741 IC (OP-AMP), OSCILLOSCOPE

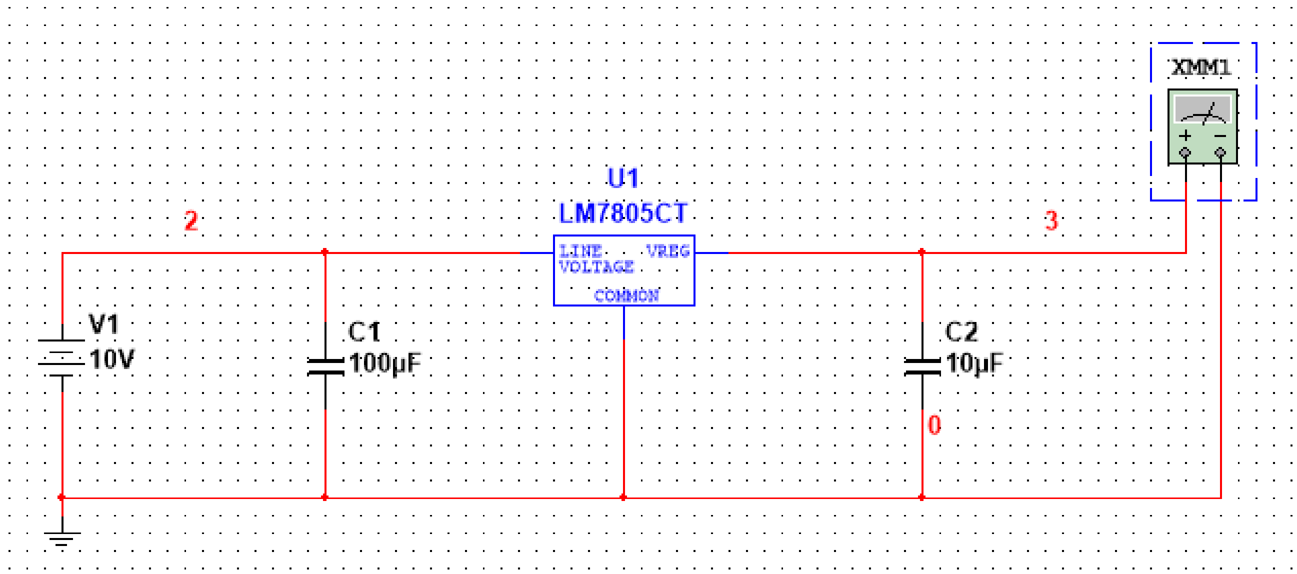


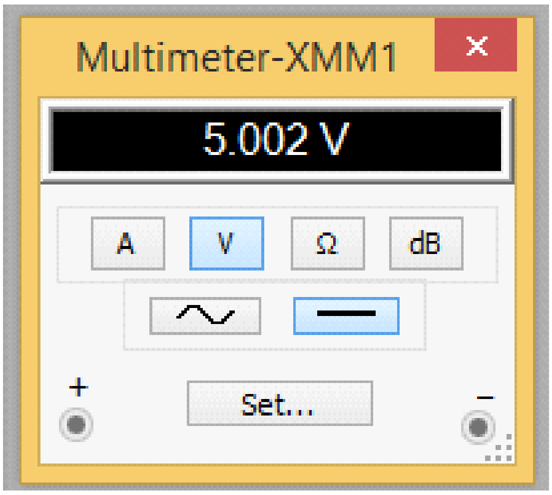


Experiment 2B: IC VOLTAGE REGULATOR

AIM: Study and simulation of IC VOLTAGE REGULATOR and observing the output voltage using MULTISIM software worksheet

COMPONENTS: DC POWER SOURCE(10V),C1(100UF),C2(10UF),MULTIMETER,LM7805CT(IC)



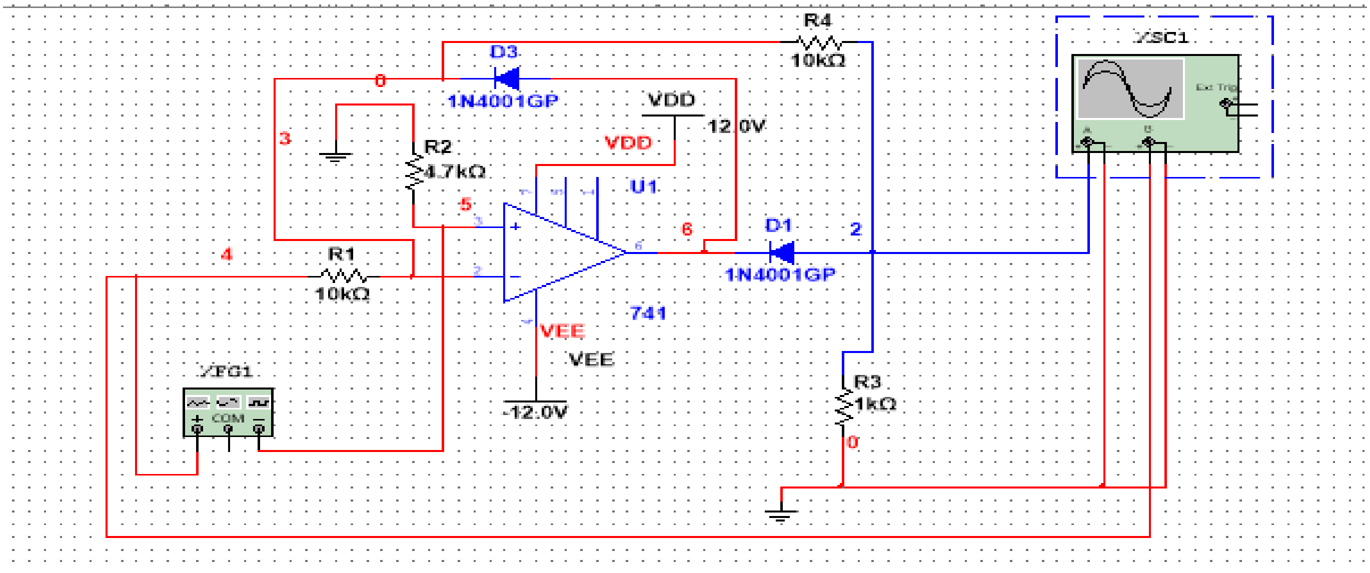


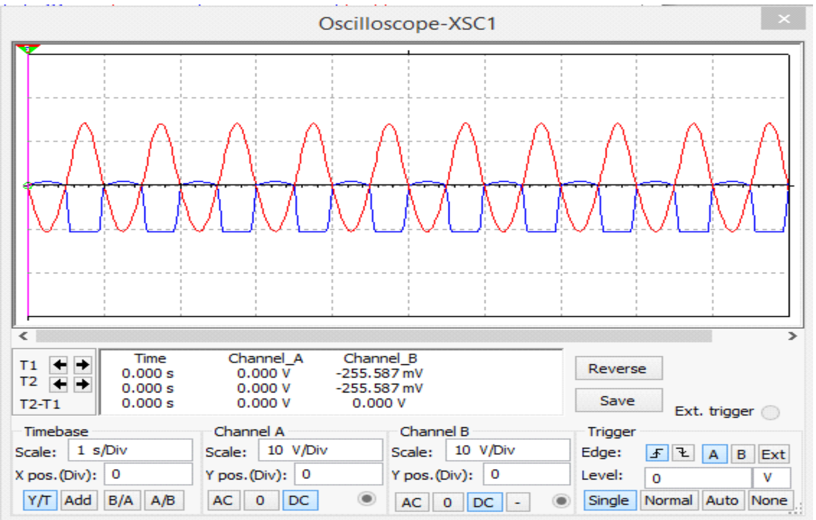
INPUT VOLTAGE	OUTPUT VOLTAGE
10V	5.02V
12V	5.02V
14V	5.02V
20V	5.02V

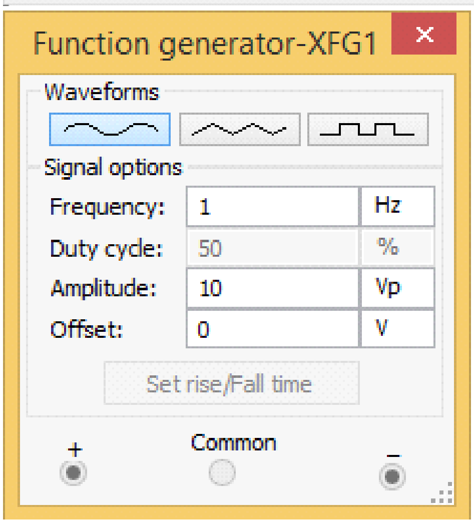
Experiment 3A: HALF WAVE PRECISION RECTIFER

AIM: Study and simulation of HALF WAVE RECTIFER and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 2 DIODE 1N4001GP, IC 741, 10K,10K, 4.7K1KOHM(RESISTOR), CRO, FUNCTION GEN



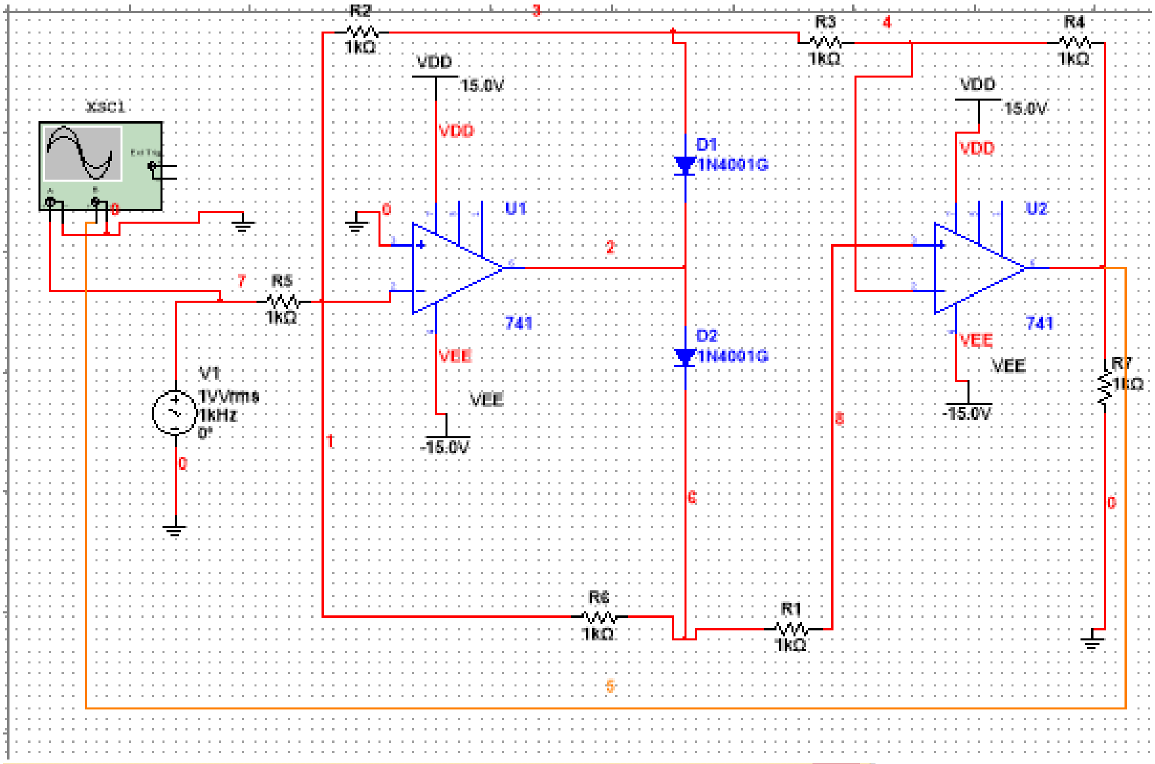


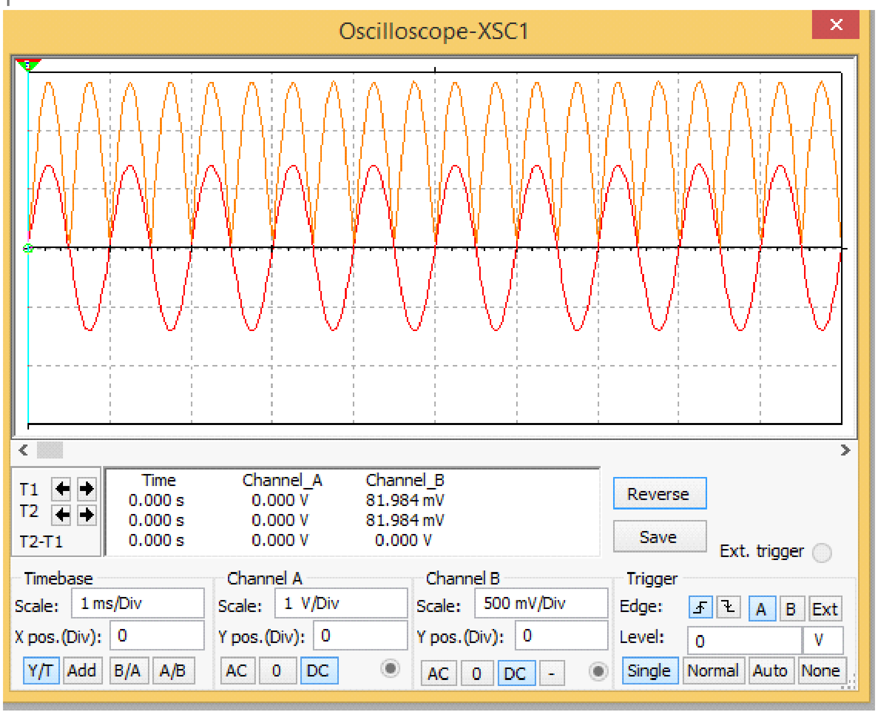


Experiment 3B: FULL WAVE PRECISION RECTIFER

AIM: Study and simulation of FULL WAVE RECTIFER and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 2 (DIODE) 1N4001GP, 2 IC 741, 1K ohm(7 RESISTOR), CRO, FUNG GEN, AC power source

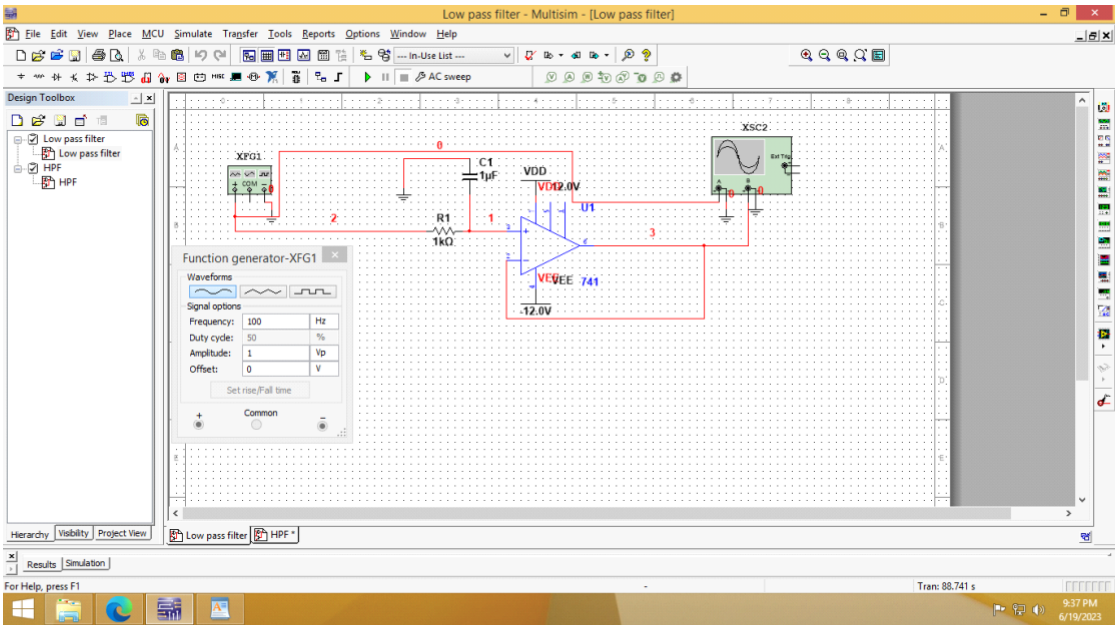


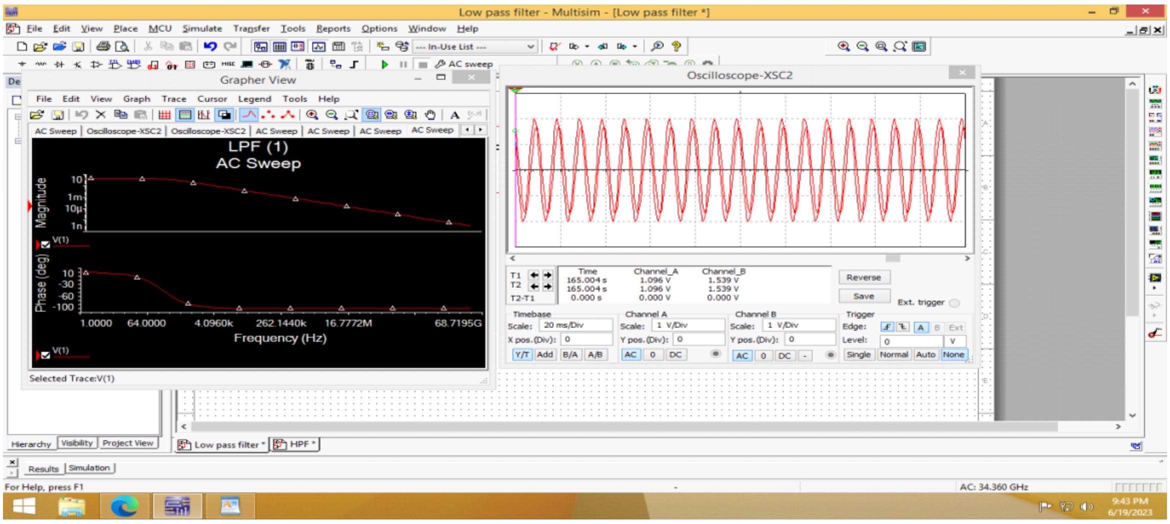


Experiment: 4A) FIRST ORDER ACTIVE Low Pass Filter

AIM: Study and simulation of First Order Low Pass filter and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 1 Micro farad (1 capacitor), IC 741, 1K ohm(1 RESISTOR), CRO, FUNG GEN





$$V_{out} \longrightarrow freq \longrightarrow f_{cutoff}\ (f_c)$$

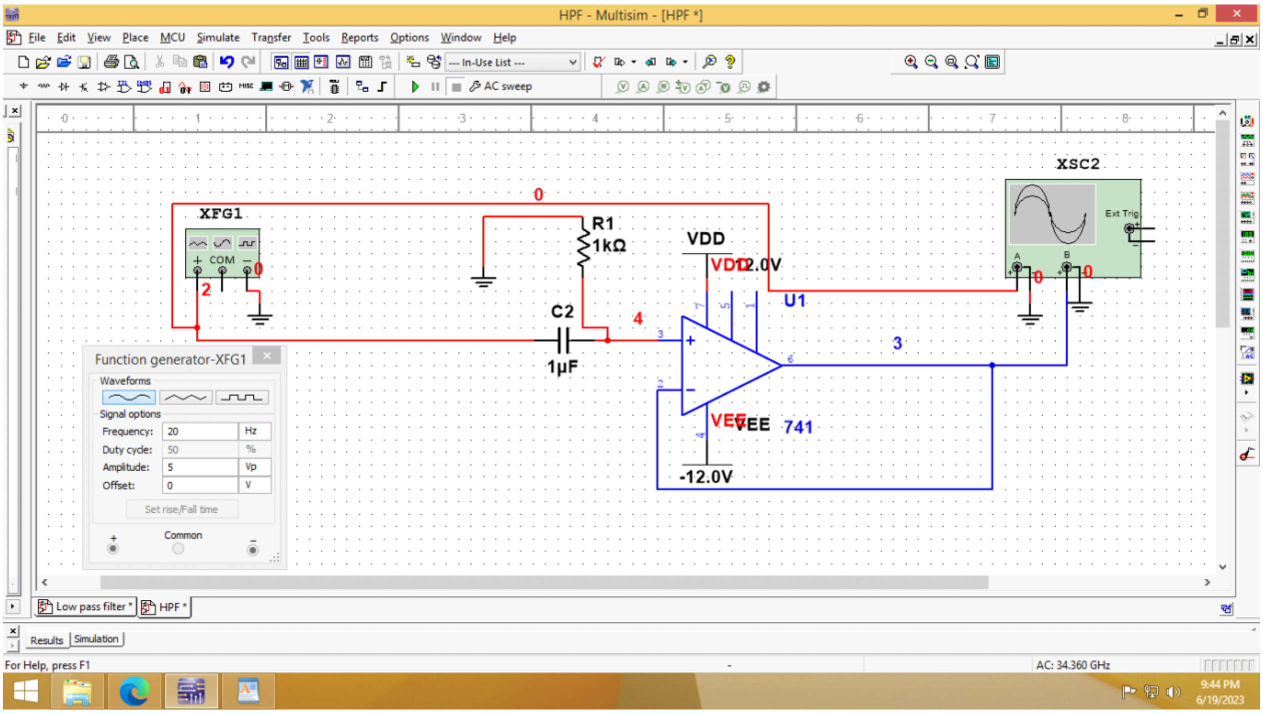
$$f_c = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\times 3{,}14 \times (1\times 10^{3}) \times (1\times 10^{-6})}$$

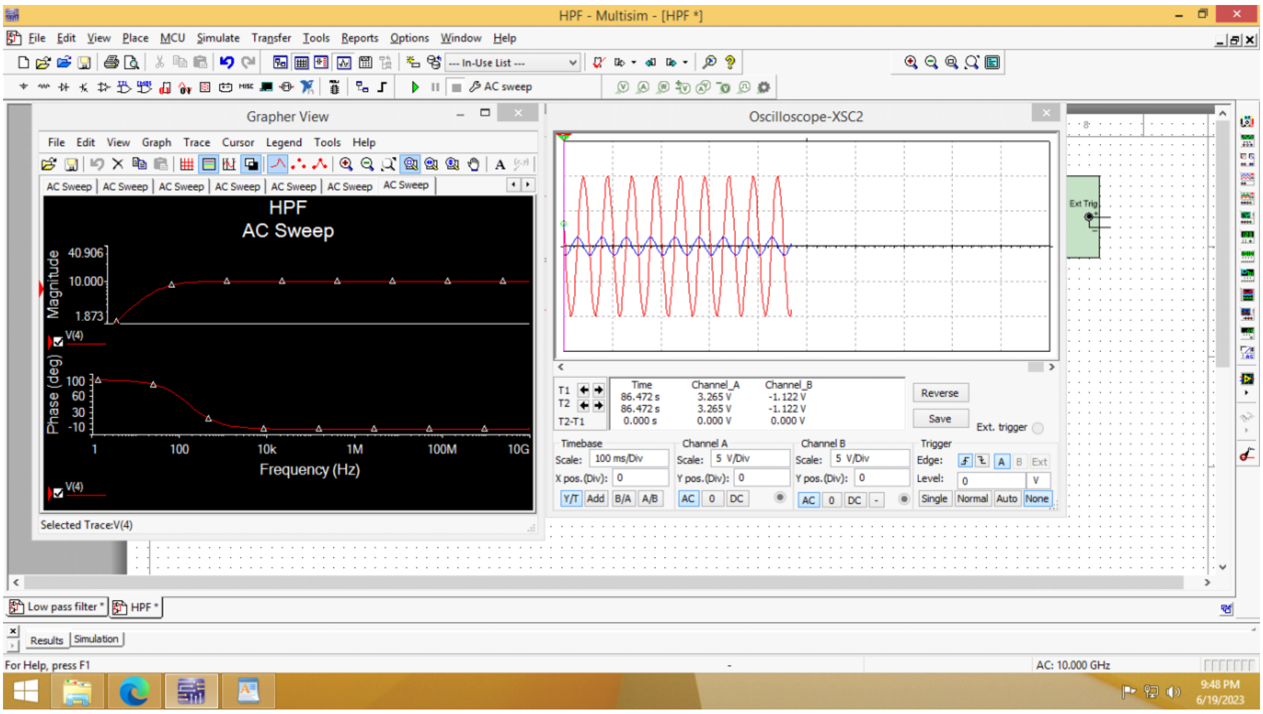
$$= \frac{1}{0{,}00628} = 159{,}24 \ Hz$$

<u>Experiment: 4B) FIRST ORDER ACTIVE High Pass Filter</u>

AIM: Study and simulation of First Order High Pass filter and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 1 Micro farad (1 capacitor), IC 741, 1K ohm(1 RESISTOR), CRO, FUNG GEN

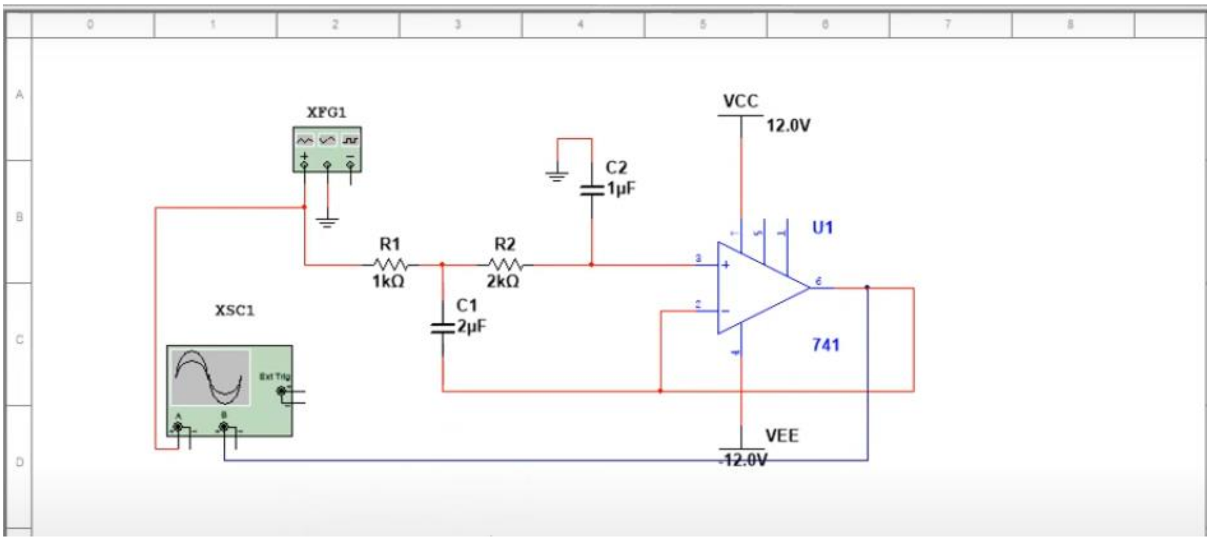




Experiment: 5A) Second ORDER ACTIVE Low Pass Filter

AIM: Study and simulation of Second Order Low Pass filter and observing the output WAVEFORM using MULTISIM software worksheet

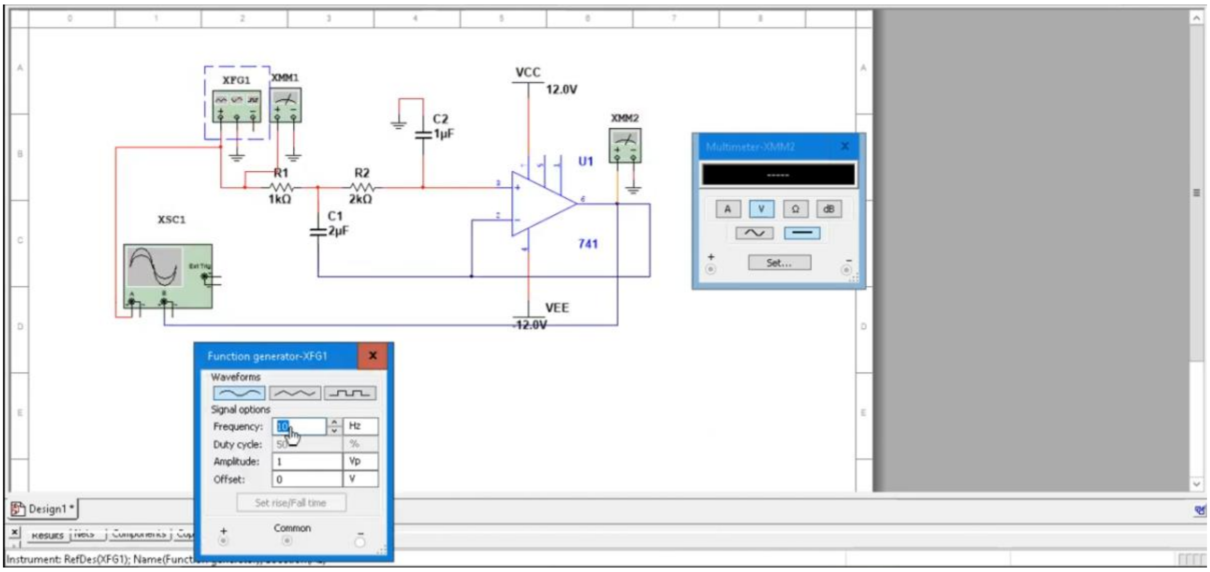
COMPONENTS: 1 Micro farad (1 capacitor), 2 Micro farad (1 capacitor), IC 741, 1K ohm(1 RESISTOR), 2K ohm(1 RESISTOR), CRO, FUNG GEN



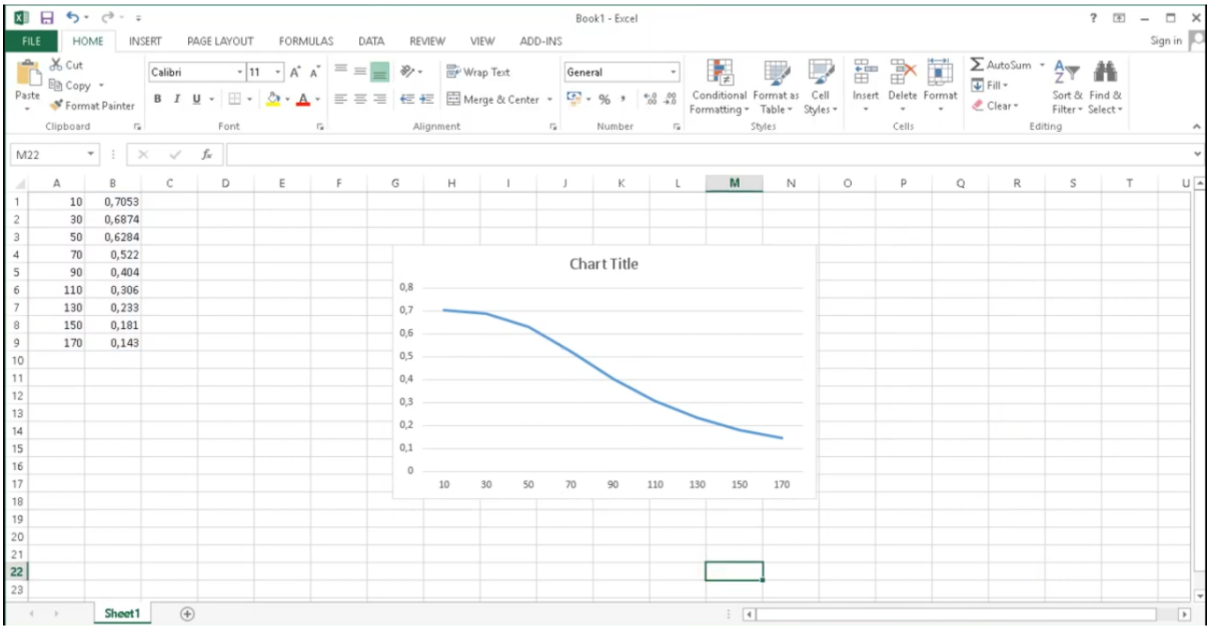
$$f_c = \frac{1}{2\pi\left(\sqrt{R_1C_1R_2C_2}\right)}$$

$$= \frac{1}{2\times 3,14\times\sqrt{(1\times 10^{3})\times(2\times 10^{-6})\times(2\times 10^{3})\times(1\times 10^{-6})}}$$

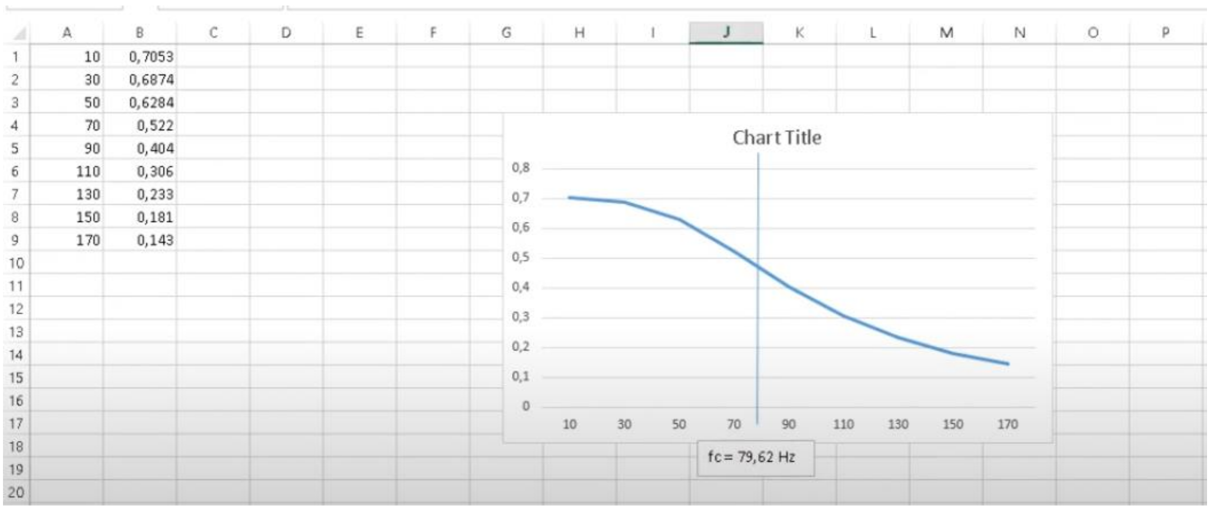
$$= \frac{1}{0,01256} = 79,62\ Hz$$



XFG1
XMM1
VCC
12.0V
C2
1µF
XMM2
R1
1kΩ
R2
2kΩ
U1
C1
2µF
XSC1
741
VEE
-12.0V
Multimeter-XMM2
Set...
Function generator-XFG1
Waveforms
Signal options
Frequency:
Hz
Duty cycle:
%
Amplitude:
1
Vp
Offset:
0
V
Set rise/Fall time
Common
Design1 *



Book1 - Excel
FILE
HOME
INSERT
PAGE LAYOUT
FORMULAS
DATA
REVIEW
VIEW
ADD-INS
M22
10 0,7053
30 0,6874
50 0,6284
70 0,522
90 0,404
110 0,306
130 0,233
150 0,181
170 0,143
Chart Title
Sheet1

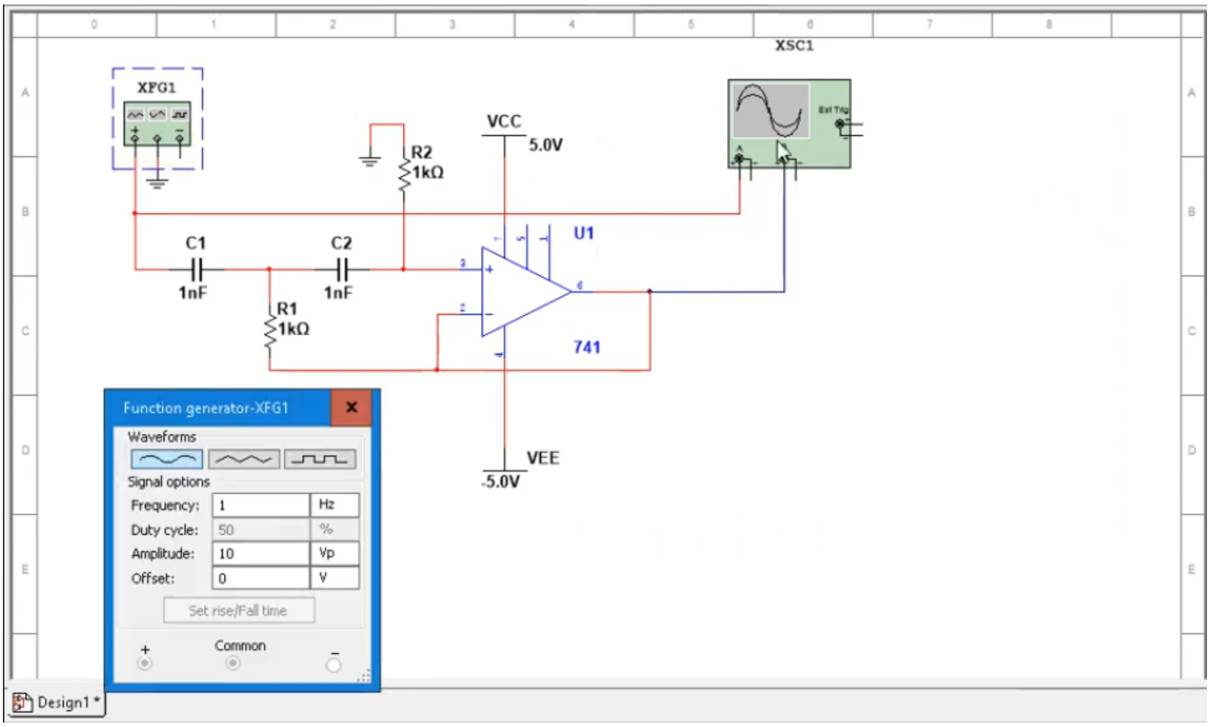


10 0,7053
30 0,6874
50 0,6284
70 0,522
90 0,404
110 0,306
130 0,233
150 0,181
170 0,143
Chart Title
fc = 79,62 Hz

Experiment: 5B) Second ORDER ACTIVE High Pass Filter

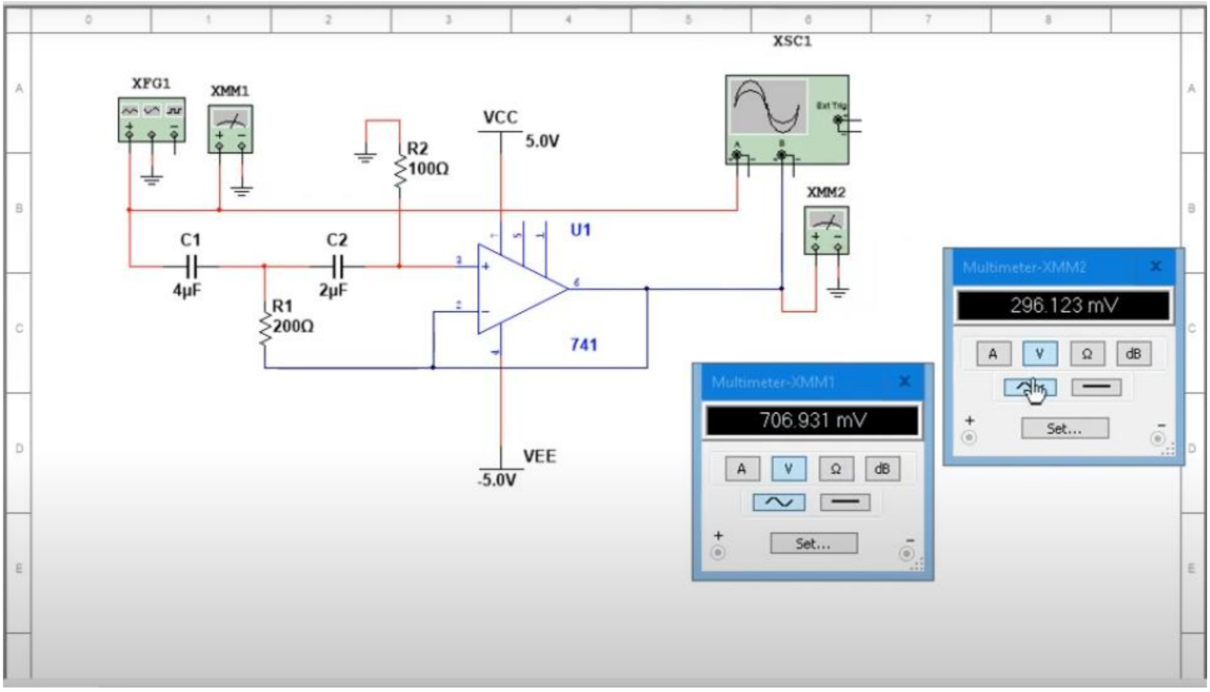
AIM: Study and simulation of Second Order High Pass filter and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 1 Nano farad (2 capacitor), IC 741, 1K ohm(2 RESISTOR), CRO, FUNG GEN

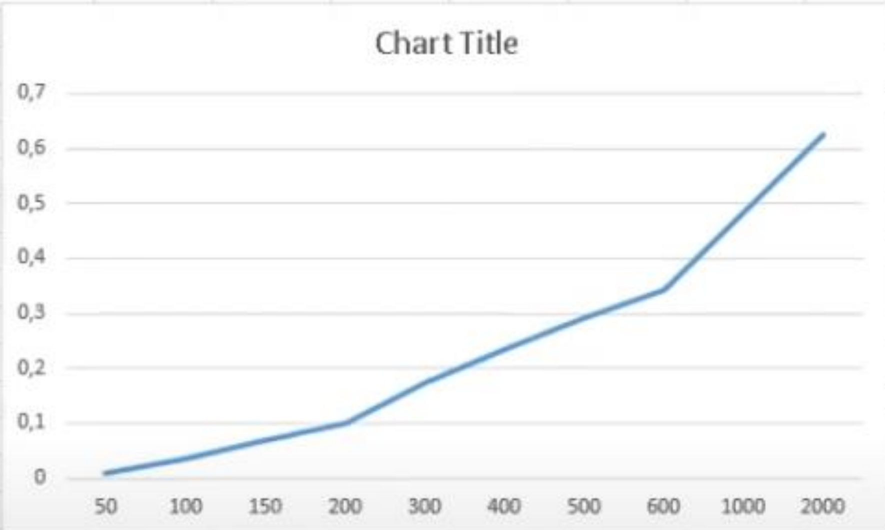


$$f_c = \frac{1}{2\pi\sqrt{R_1C_1R_2C_2}} = \frac{1}{2.3{,}14.\sqrt{200.4.10^{-6}.100.2.10^{-6}}}$$

$$= \frac{1}{0{,}002512} = \underline{398{,}09\ Hz}$$



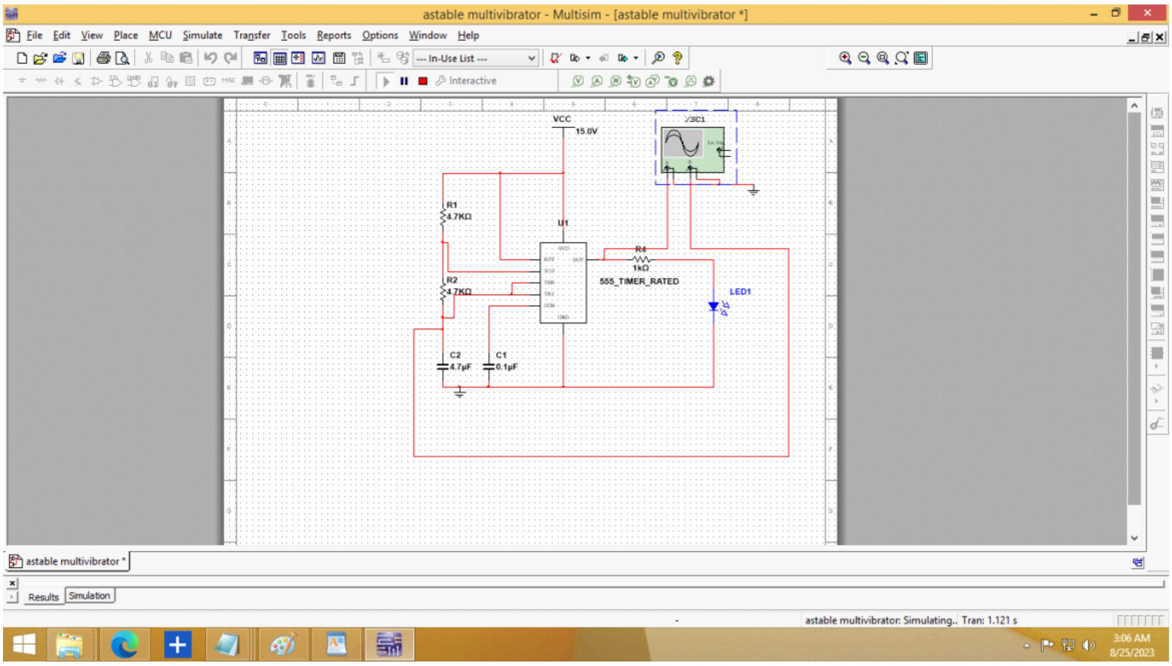
fre	vo
50	0,01
100	0,037
150	0,07
200	0,1
300	0,175
400	0,236
500	0,292
600	0,342
1000	0,484
2000	0,625



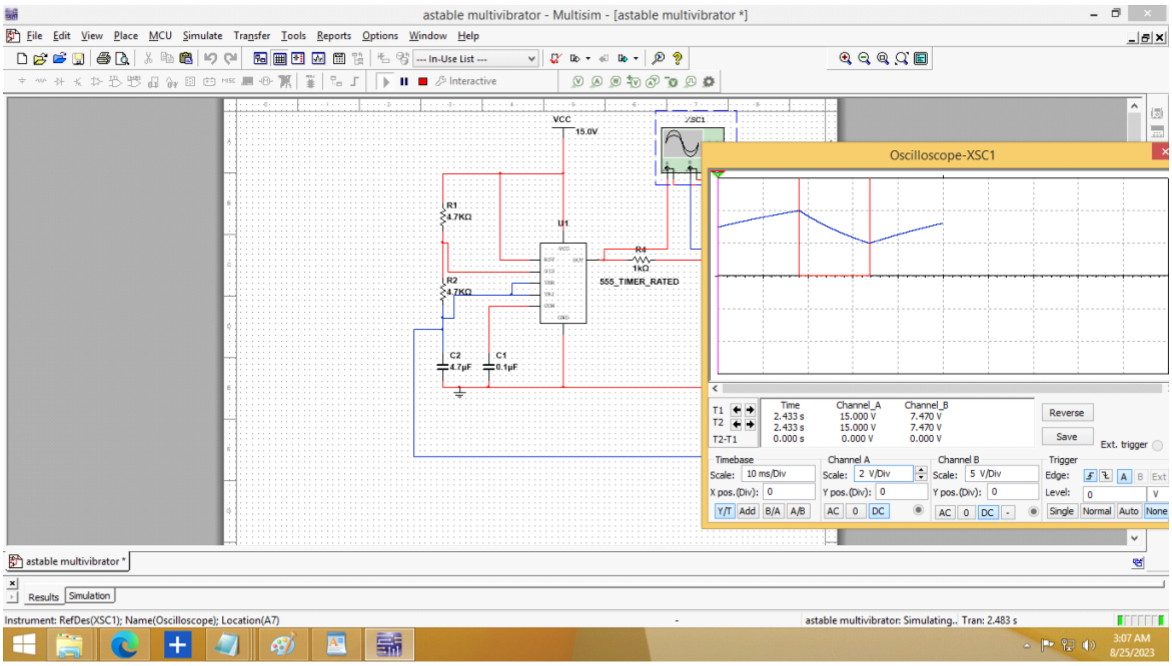
Experiment: 6 Astable Multivibrator

AIM: To design the astable multivibrator using IC555 timer and to find out the ON and OFF time

COMPONENTS: 555 timer IC , resistor , capacitor ,Oscilloscpe , LED.



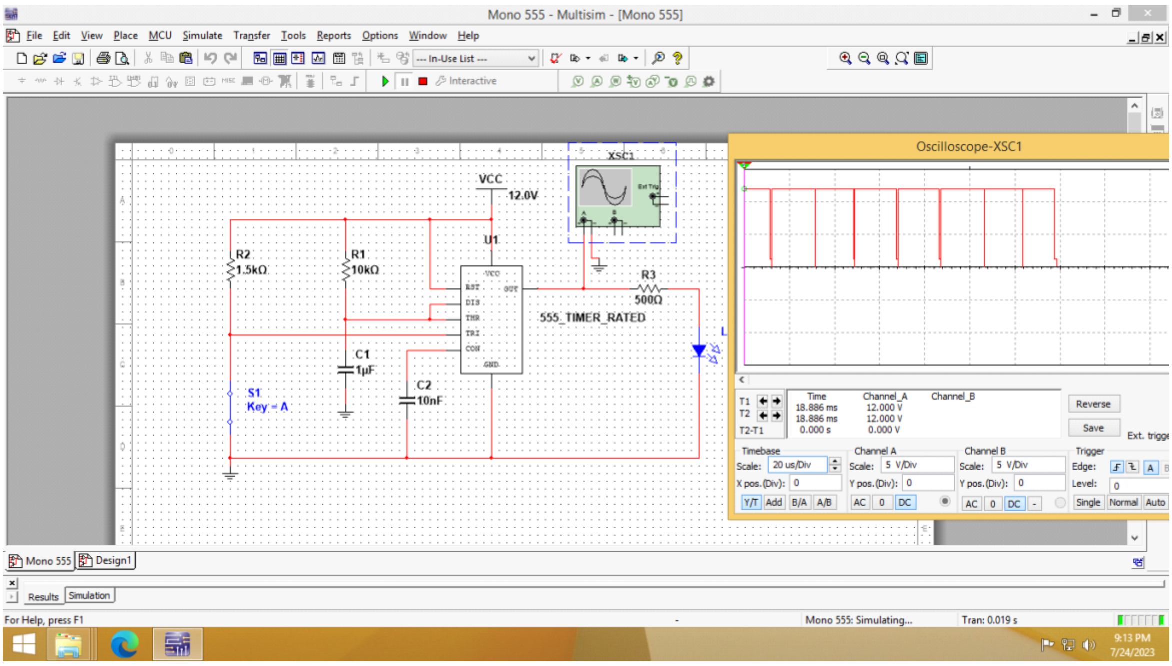
Design:



Experiment :7   Monostable Multivibrator

AIM: To design the monostable multivibrator using IC555 timer and to find out the ON and OFF time

COMPONENTS: 555 timer IC , resistor , capacitor ,Oscilloscpe , LED.



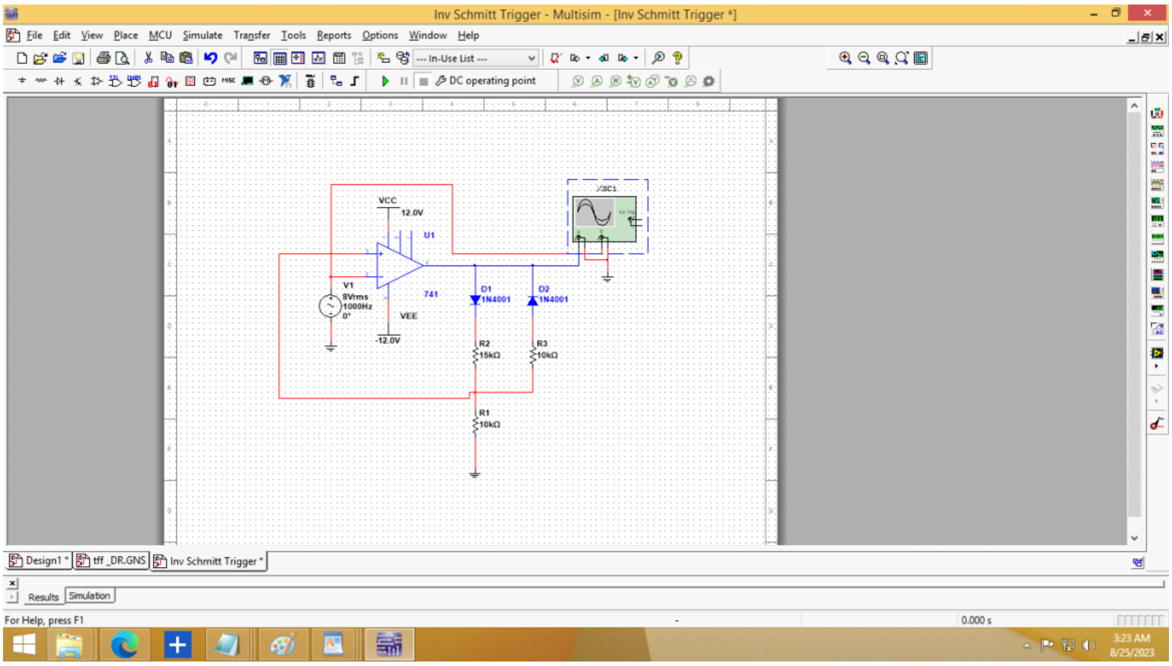
Design: $T=1.1R_1C_1$

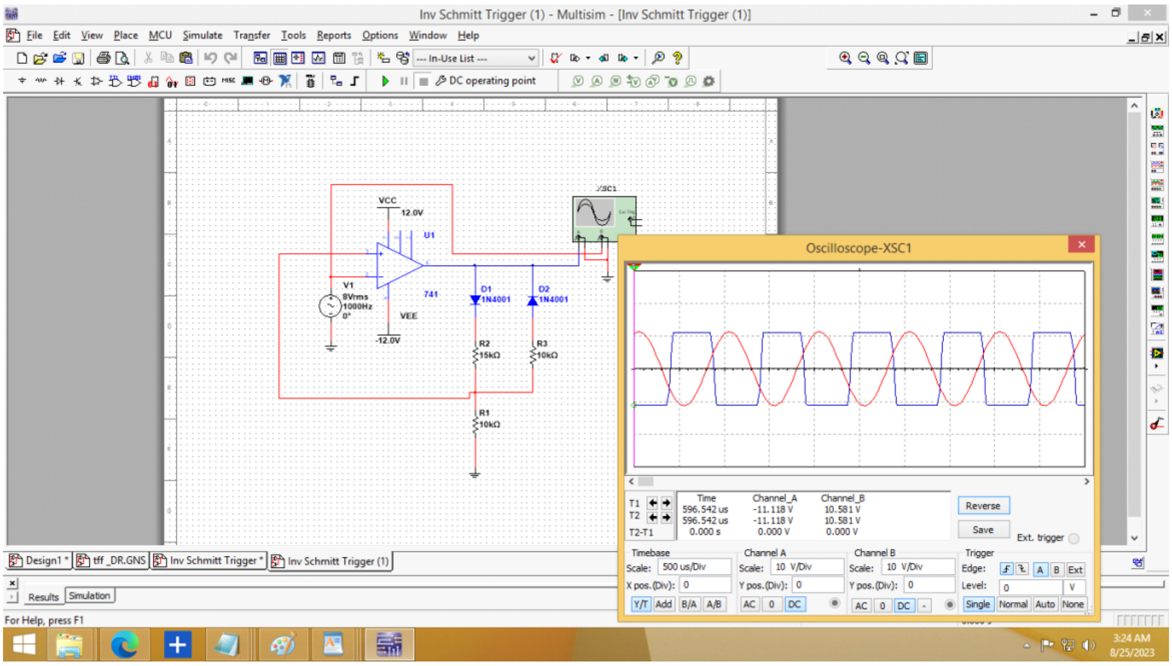
Let R1=10kΩ; C1=1µF ………T=11ms

Experiment :8) Inverting Schmitt Trigger

AIM: To design Inverting Schmitt Trigger using 741 and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 741 op-amp , resistor , capacitor ,Oscilloscpe , LED.



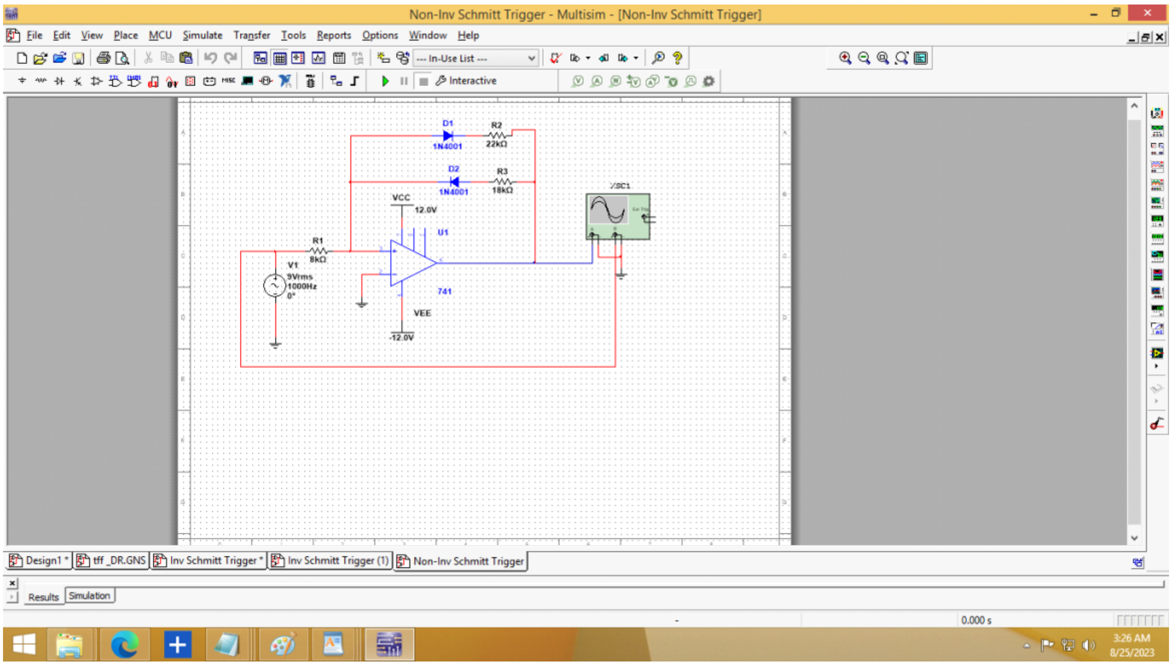


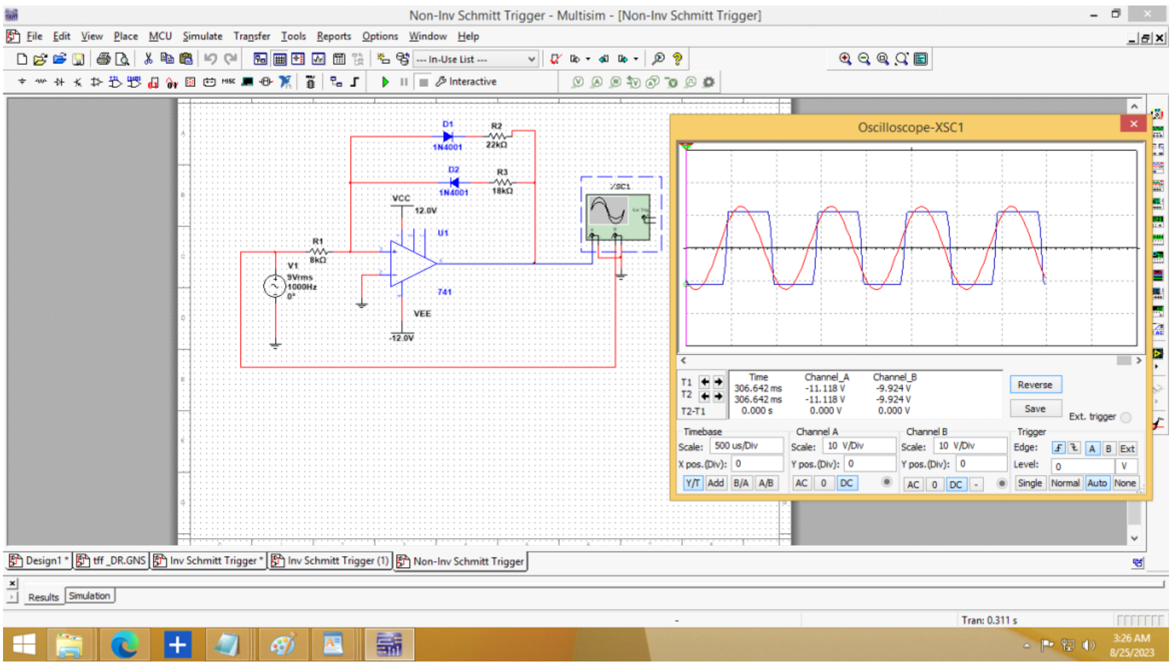
Design given separately

Experiment :9) Non-Inverting Schmitt Trigger

AIM: To design Non Inverting Schmitt Trigger using 741 and observing the output WAVEFORM using MULTISIM software worksheet

COMPONENTS: 741 op-amp , resistor , capacitor ,Oscilloscpe , LED.



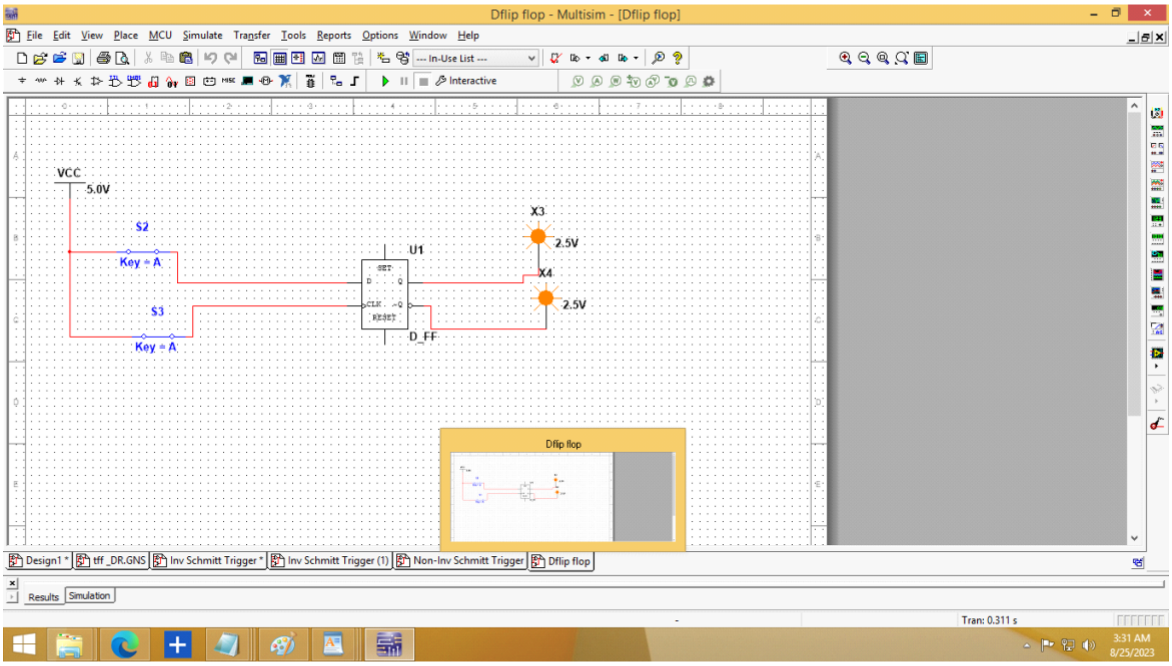


Design given separately

Experiment :10) D-Flip Flop Realization

AIM: To design D-Flip Flop and verify the output with truth table

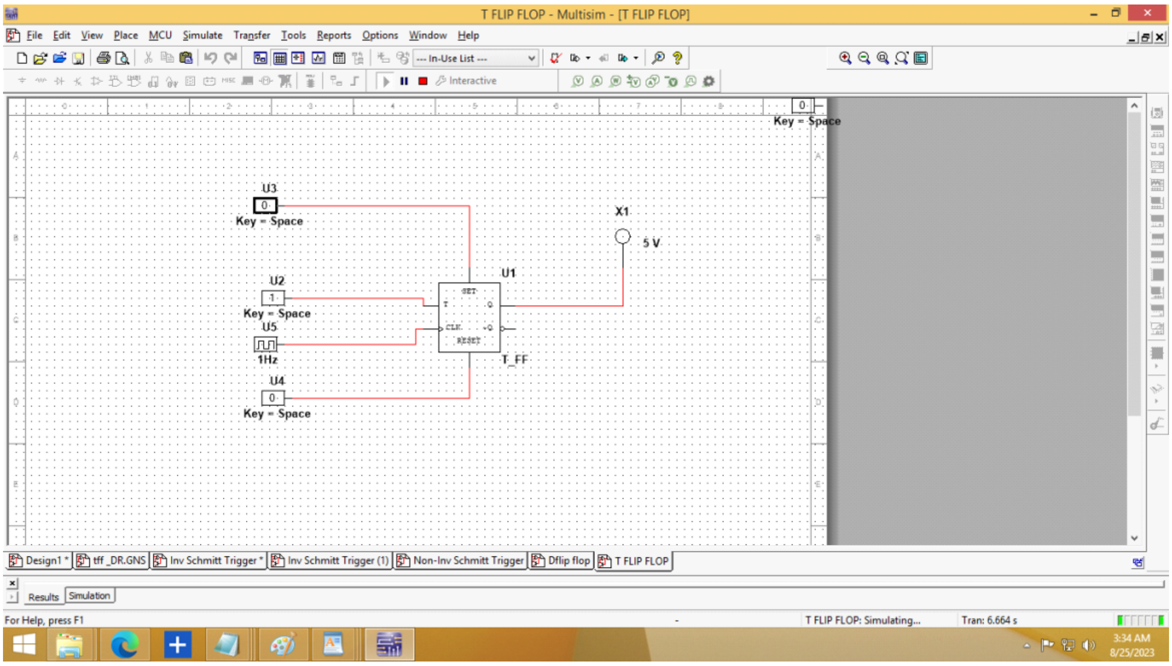
COMPONENTS: D-Flip Flop , Vcc, Switch ,Probe.



Experiment :11) T-Flip Flop Realization

AIM: To design T-Flip Flop and verify the output with Case.

COMPONENTS: T-Flip Flop , Digital clock ,Interactive Digital Constant.

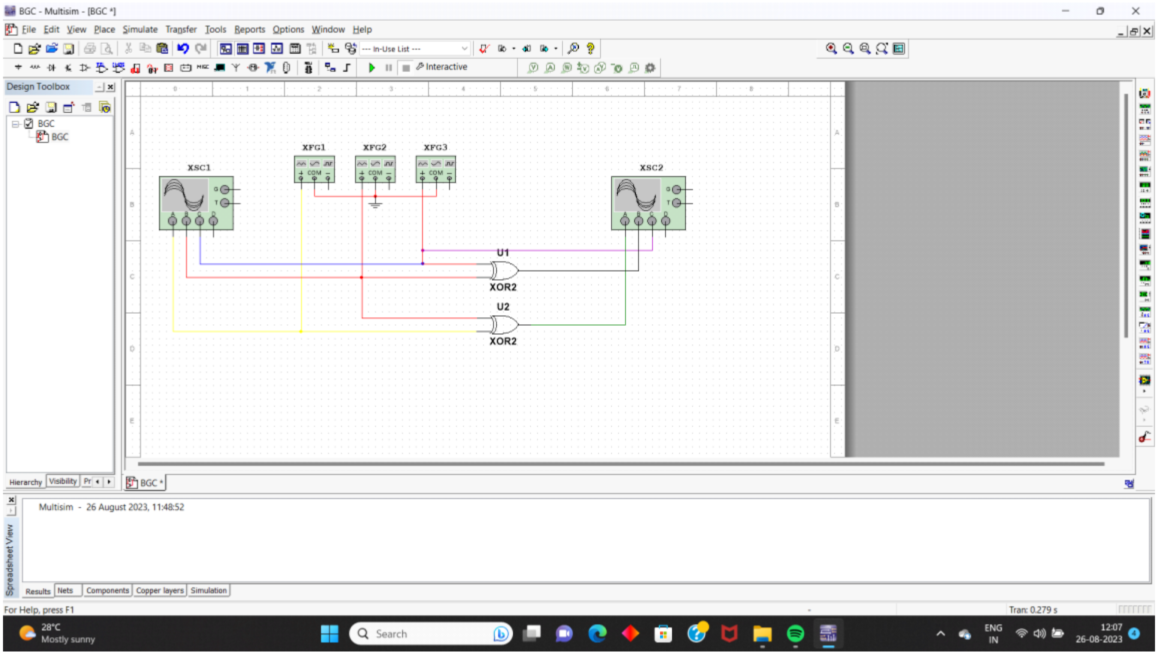


Experiment 12: Code Conversions

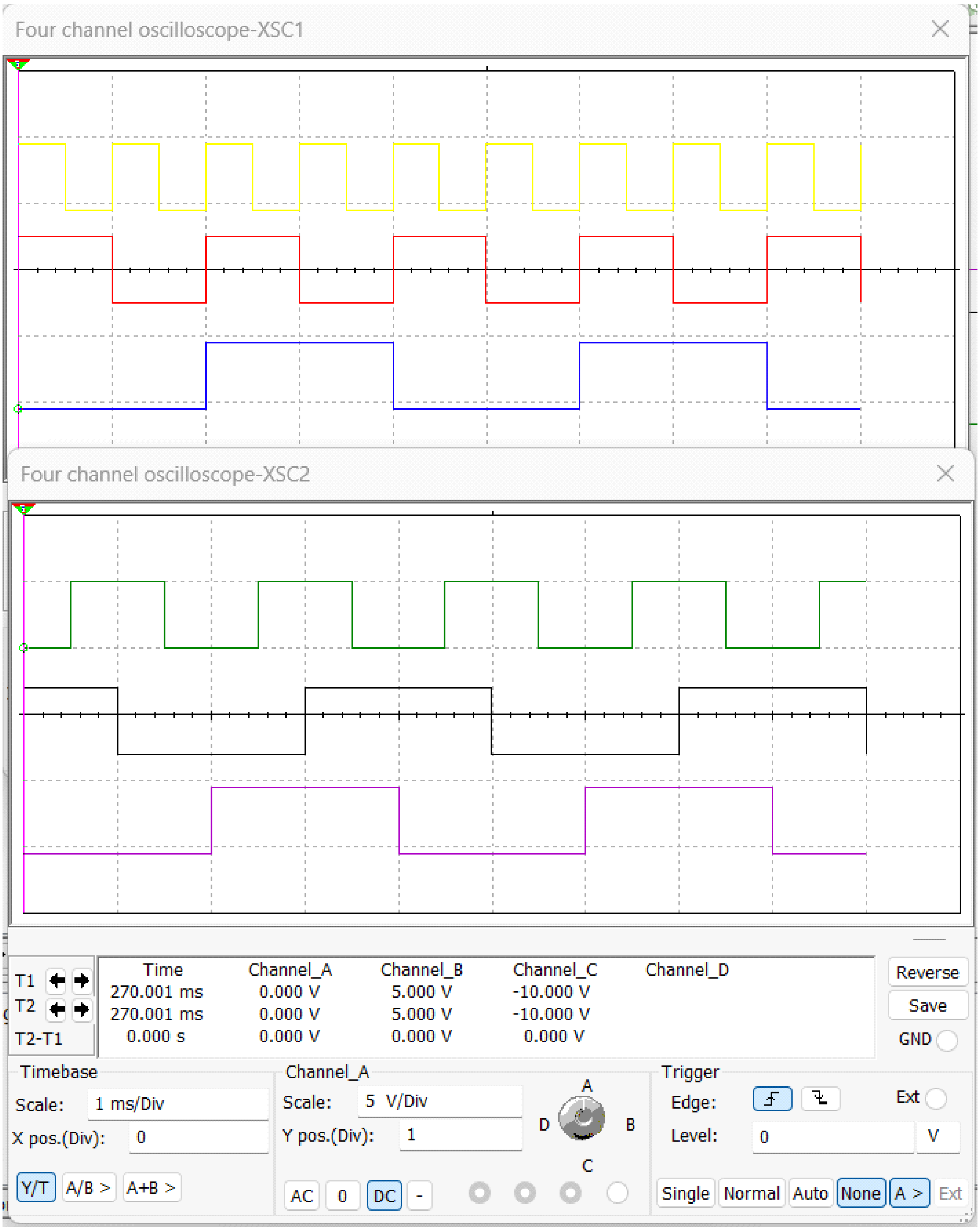
AIM: To design code converter and verify the output for binary to Gray code conversion.

COMPONENTS: Oscilloscope , Function Generator ,Op Amp 741.

Circuit Diagram



Waveforms



Four channel oscilloscope-XSC1
Four channel oscilloscope-XSC2
Time Channel_A Channel_B Channel_C Channel_D
T1 270.001 ms 0.000 V 5.000 V -10.000 V
T2 270.001 ms 0.000 V 5.000 V -10.000 V
T2-T1 0.000 s 0.000 V 0.000 V 0.000 V
Reverse
Save
GND
Timebase
Scale: 1 ms/Div
X pos.(Div): 0
Y/T A/B > A+B >
Channel_A
Scale: 5 V/Div
Y pos.(Div): 1
AC 0 DC -
A B C D
Trigger
Edge:
Ext
Level: 0 V
Single Normal Auto None A > Ext